

Explaining Legal Moves in Equations and Inequalities

Answer Key

Algebra 1

Quick Answers

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|------------------------------|---|
| 1. 7 | 6. 35 |
| 2. (a) = —, (b) = —, (c) = 5 | 7. (a) solution set = $(-\infty, -3]$, (b) = — |
| 3. 8 | 8. (a) = —, (b) = 4 |
| 4. $(-4, \infty)$ | 9. $(-\infty, -8)$ |
| 5. (a) = —, (b) = 6 | 10. (a) = 11, (b) = — |
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What is verified

10 of 16 answers machine-checked against SymPy, across 10 problems.

— marks an answer only you can judge — problems 2, 5, 7, 8, 10. The worked solution states what a correct response must contain.

The author marked 2 blanks as working space (*not* answers — nothing checks what lands there).

Curriculum

Level: Algebra 1

HSA-REI.A.1 — *problems 2, 5, 7, 8, 10*

HSA-REI.B.3 — *problems 1, 2, 3, 4, 5, 6, 7, 8, 9, 10*

Difficulty 1–5 of 5 across 16 tagged checks.

1. Balance: left pan $x + 4$ grams, right pan 11 grams. Complete the model and give x .

Solution: Removing 4 grams from both pans leaves x on the left and 7 on the right, and the balance is still level — that is exactly what the subtraction property of equality says. The completed model row reads x and 7.

$x = 7$

2. $3x - 7 = 14 \rightarrow 3x = 21 \rightarrow x = 7$: (a) name the first move; (b) name the second; (c) solve $5x + 2 = 27$.

Solution (a) — instructor-judged. 7 was added to both sides: the addition property of equality. On the left $-7 + 7$ leaves $3x$; on the right $14 + 7 = 21$. Full credit states the operation *and* that both sides received it; naming only the number 7 does not.

Solution (b) — instructor-judged. Both sides were divided by 3 (equivalently, multiplied by $\frac{1}{3}$): the division property of equality. Full credit again names the operation and both sides.

Solution (c): Subtract 2 from both sides: $5x = 25$. Divide both sides by 5.

$x = 5$

3. Solve $4(x - 3) = 20$.

Solution: Route one, divide first: dividing both sides by 4 gives $x - 3 = 5$, then adding 3 to both sides gives $x = 8$. Route two, distribute first: $4x - 12 = 20$, add 12 to both sides for $4x = 32$, divide both sides by 4. Both routes are legal because each move either does the same thing to both sides or rewrites one side into an equal expression. $x = 8$

4. Solve $-3x < 12$.

Solution: Divide both sides by -3 . Dividing by a negative reverses the order of the two sides, so the $<$ becomes $>$: $x > -4$. Checking with a value from the answer set, $x = 0$ gives $0 < 12$, which is true; and $x = -5$ gives $15 < 12$, which is false — so the direction is right. $x > -4$

5. Student work $2x + 5 = 3x - 1 \rightarrow 5 = x - 1 \rightarrow 6 = x$: (a) name the moves; (b) solve it.

Solution (a) — instructor-judged. Step one: $2x$ was subtracted from both sides (subtraction property of equality), leaving $5 = x - 1$. Step two: 1 was added to both sides (addition property), leaving $6 = x$. Both moves are legal and the student's work is correct. Full credit names both moves and says each was applied to both sides.

Solution (b): Subtracting $2x$ and then adding 1 to both sides gives $6 = x$, which is the same statement as $x = 6$. $x = 6$

6. Solve $\frac{x}{5} - 3 = 4$.

Solution: Add 3 to both sides: $\frac{x}{5} = 7$. Then multiply both sides by 5: $x = 35$. Check: $35 \div 5 - 3 = 7 - 3 = 4$. $x = 35$

7. $5 - 2x \geq 11$: (a) solve; (b) does the sign change, and where?

Solution (a): Subtract 5 from both sides: $-2x \geq 6$. Divide both sides by -2 , which reverses the sign: $x \leq -3$. $x \leq -3$

Solution (b) — instructor-judged. Yes, it changes, and at exactly one step: dividing both sides by -2 . Subtracting 5 from both sides does not affect the direction, because shifting both sides by the same amount preserves which one is larger; multiplying or dividing by a negative swaps which one is larger, so the symbol must be reversed to keep the statement true. Full credit names that step as the reason; a bare yes or no earns half credit.

8. $3(x + 2) = 18 \rightarrow 3x + 2 = 18 \rightarrow \dots$: (a) which step is illegal; (b) solve correctly.

Solution (a) — instructor-judged. The very first step. $3(x + 2)$ was rewritten as $3x + 2$: the factor 3 reached the x but not the 2. The distributive property requires it to multiply every term inside, giving $3x + 6$. Note that this is not a property-of-equality error at all — nothing was done to both sides — it is a bad rewrite of one side, which is just as fatal. Full credit points at that first line.

Solution (b): $3(x+2) = 18$ becomes $3x+6 = 18$; subtract 6 from both sides for $3x = 12$; divide both sides by 3. $x = 4$

9. Solve $-\frac{x}{4} + 1 > 3$.

Solution: Subtract 1 from both sides: $-\frac{x}{4} > 2$. Now multiply both sides by -4 ; the multiplier is negative, so the sign reverses: $x < -8$. Check with $x = -12$: $3 + 1 = 4 > 3$, true. $x < -8$

10. $\frac{2x-1}{3} + 4 = x$: (a) solve; (b) justify each move.

Solution (a):

$\frac{2x-1}{3} + 4 = x$	
$2x - 1 + 12 = 3x$	multiply both sides by 3
$2x + 11 = 3x$	combine like terms on the left
$11 = x$	subtract $2x$ from both sides

Check: $\frac{2(11)-1}{3} + 4 = \frac{21}{3} + 4 = 7 + 4 = 11$.

$x = 11$

Solution (b) — instructor-judged. Multiplying both sides by 3 is the multiplication property of equality — a both-sides move. Combining $-1 + 12$ into 11 is a rewrite of the left side only, justified by arithmetic, not by any property of equality. Subtracting $2x$ from both sides is the subtraction property — another both-sides move. Full credit distinguishes the both-sides moves from the one-side rewrite.